



ChipSat V5 (2022-2025)

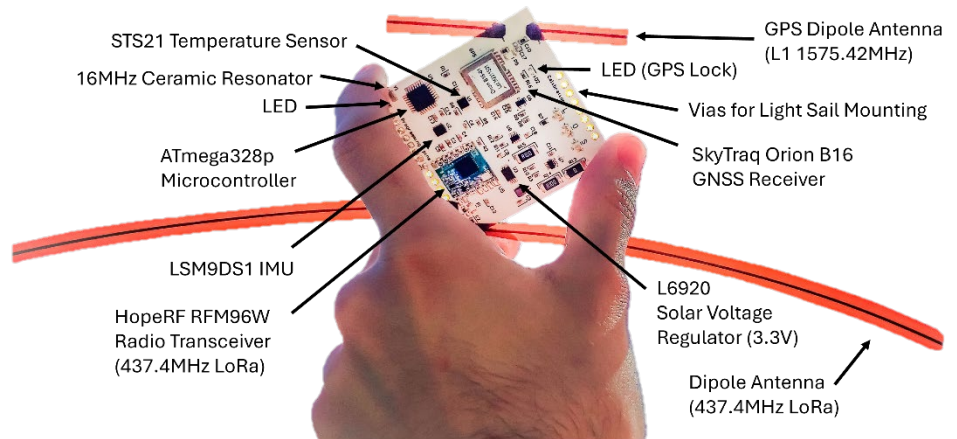
Flight-proven LEO ChipSat

Designed for free-flying light sails
(Alpha CubeSat mission)

Applications

- Low power, long range communications
- Remote sensing / distributed sensing
- Short-duration multi-day LEO operation
- Atmospheric or ground-based sensing
- Telemetry for lightweight solar/laser sails

Dimensions: 58mm x 58mm x 3mm | **Mass:** 6g



ChipSat layout

Capabilities Summary

Power: Wireless operation via 780mW PV (one-sided); **Potential modifications:** dual-sided solar or battery-only

Processing: 16MHz CPU speed; 32KB Flash, 2KB SRAM; **Potential modification:** external flash memory

Comms: Semtech SX1276; +20dBm TX; 70cm Band; LoRa, (G)FSK, (G)MSK, and OOK

Position/Attitude determination: GNSS Receiver + 9-DOF IMU (accelerometer, gyroscope, magnetometer)

Environmental Sensing: Temperature + 3-DOF magnetometer from IMU

Structure: Thin flexible polyimide (Kapton) PCB

Survivability

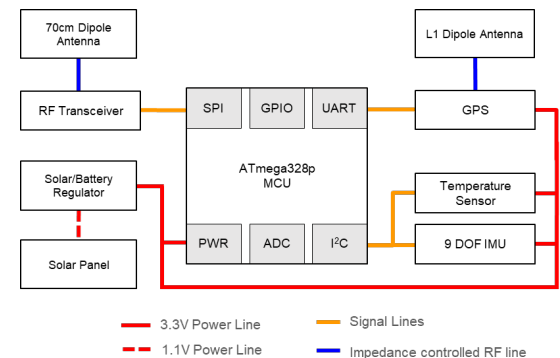
Vacuum ☒ | Launch loads/vibration ☒ | Radiation* ☒ | Terminal-velocity ground impact ☒

Operating temperature: -40°C – +85°C (additional limitations if modified to include energy storage)

*No radiation shielding / rad-hard avionics –recommended for short-term missions w/ large quantities of ChipSats

Additional Specifications

MCU	ATmega328p-AU						
I/O Pins:	UART x1, I2C x1, SPI x1, 10-bit ADC x6, GPIO x8				Sleep current:		4.4 μ A
Clock Speed:	16MHz	Flash Memory:	32KB	SRAM:	2KB	EEPROM:	1KB
Sensors	STS21 Temperature Sensor			LSM9DS1 9-DOF IMU			
	Temp			Accel		Gyro	Mag
Resolution:	0.08°C			0.732 mg		0.875 dps	0.014 μ T
Range:	-40° - 125°C			$\pm$ 16g		$\pm$ 245 dps	$\pm$ 400 μ T
GPS	SkyTraq Orion B16-01; L1 GPS/Galileo/GLONASS/BeiDou/QZSS; Space-rated						
Accuracy:	Position:	2m	Velocity:	0.1m/s	Time:	5ns	TTFF: 1 min
Power	L6920 regulator IC + 780mW GaAs PV No energy storage onboard						
Lifespan	Indefinite operation under solar illumination (dependent on sun-orientation)						
Comms	SX1276 (built into HopeRF RFM96W); also capable of (G)FSK, (G)MSK, OOK						
Freq:	433MHz	BW:	7.8 - 500KHz	TX PWR:	+20dBm	Modulation:	LoRa
Approx. Range:	(w/ below receiver equipment)			50km	LEO	GEO	Cislunar
Estimated LoRa Data rate (bps):		Low-gain omni		488 bps	146 bps	X	X
		High-gain Yagi		3.5 kbps	488 bps	45 bps	X
		Large parabolic dish		22 kbps	3.5 kbps	146 bps	11 bps



ChipSat architecture

Est. cost (per ChipSat):

\$1300 USD (Orbital) / \$200 USD (Terrestrial)

For more information:

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